Precision Partner

Enhancing GFRC craftsmanship with industry 4.0 factory-floor feedback

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This paper presents a novel human-machine collaborative approach to automatic quality-control of Glass-Fiber Reinforced Concrete (GFRC) molds directly on the factory floor. The framework introduces Industry 4.0 technologies to enhance the ability of skilled craftsmen to make molds through the provision of horizontal feedback regarding dimensional tolerances. Where digital tools are seldom used in the fabrication of GFRC molds, and expert craftsmen are not digital experts, our implementation of automated registration and feedback processes enables craftsmen to be integrated into and gain value from the digital production chain. In this paper, we describe the in-progress framework, Precision Partner, which connects 3d scanning and point cloud registration of geometrically complex and varied one off elements to factory floor dimensional feedback. We firstly introduce the production context of GFRC molds, as well as industry standards for production feedback. We then detail our methods, and report the results of a case study that tests the framework on the case of a balcony element.

Keywords: *3d Scanning, GFRC, Feedback, Automation, Human in the loop, Digital Chain*

INTRODUCTION

The construction Industry is the second least digitized in the world [1]. Despite the increasing ubiquity of digital tools within design, simulation and management workflows (Claypool et al. 2019), most fabrication, assembly and construction practices rely on hand-work and craft expertise, whether in factories or on-site. This gap becomes relevant in the case of projects that involve complex geometries, in which dimensional feedback and quality control is central. While automated quality-control processes have been long established in production lines (Matt et al. 2020) to inspect for faults in high-volume production, implementing registration and feedback processes in hand craft shop-floor environments, the typical mode of production in the building industry, to inspect geometrically complex one-off elements remains a challenge.

Adapting Industry 4.0 technology to the factory floor would allow for skilled hand-craft processes to

be integrated with, and gain value from, the digital information chain. Skilled craftsmen perform central and critical roles in many manufacturing and construction processes, particularly within manufacturing tasks where the level of complexity and variation makes automation highly difficult or inefficient (Autor 2015). Within this type of manufacturing activity, human craft skills are able to contribute dexterity, flexibility, sensory and decision-making ability [2], and are recognised to outperform current autonomous and automated systems (Autor 2015). Because of this, it is important to understand how current Industry 4.0 research can extend beyond its current focus of establishing connections between machines and systems, and to consider the human in the loop. In this paper we explore how Industry 4.0 can enable new types of interface between craft expertise and digital design information through the idea of digital assistance, and the proposition that coupled human-automated activities can assist expert craftsmen in becoming even more effective and efficient.

In this paper, we describe a framework for automating feedback processes from 3d scanning to assist craftsmen in evaluating the geometric quality of their work. We contextualise our research through description of a specific manufacturing process - GFRC manufacturing - and against analysis of state of the art in production feedback. We introduce a new, prototypical framework in the context of expert mold makers creating molds for precast fiber-reinforced concrete building elements. Through feedback from 3d scanning, this framework enables craftsmen to assess their production against digitally established critical dimensions at any point throughout the making process, and to make spot measures that they would not be able to perform via physical measuring techniques. The prototypical framework aims towards a seamless solution, with the objective that automation should reduce any disturbance of the making process to a minimum, so that gaining feedback does not require expert knowledge. Key aspects of this ambition are the development of registration protocols and the creation of a customised interface. The framework is developed and evaluated in an interdisciplinary collaboration of researchers from CITA, GFRC producer BBFiberbeton and the metrology provider TickCAD.

MANUFACTURING ARCHITECTURAL GFRC ELEMENTS

Within architecture, GFRC is used to manufacture lightweight, prefabricated cladding elements, facade panels, and balconies with varying forms and shapes [3]. GFRC elements can be categorised along a scale of increasing geometric and manufacturing complexity as either rectilinear, faceted and free-form, and their manufacture requires a series of stages to be connected together. The typical process of manufacture starts with an analysis of the client's input data (2d CAD or 3d BIM), detailing and optimisation of the design according to budget, manufacturing and other constraints and the definition of quality criteria for the later element, including surface quality and tolerances in dimensions.

In a second stage - process planning - the mold is designed and drawings of its parts are generated. Most molds are made from flat sheet materials, combining wood and plastic, cut to shape (Fig. 1). Because of the high variability in GFRC element design, and resultant high geometric variability design, making and assembling of molds is not automatised and the resulting quality relies on the skills of craftsmen. In order to effect overall cost and efficiency the design of GFRC elements and molds is tweaked, so that a mold can be reused many times or easily adapted to changing geometries in the production of an element.

The 25% of molds at our industry partner, which are faceted or free-form are more challenging, as 2D drawings can fail to describe the complexity of the form regardless of the number of details included. Here, CNC routing can be used for the production of curvilinear sub elements of molds or the 3d milling of entire mold surfaces (Kreysler, 2017).

Finally GFRC is sprayed in layers into the mold and rolled a handheld roller (Fig.1), which requires high level of craftsmanship again, since the strength of the concrete is determined from the correct mixture and compacting the concrete slurry with the fibres (Brandt 2020).

Quality Control is imperative at all stages of production. At two stages geometric discrepancies between the built mold and its specification are measured: 1) during the manufacturing of the mold and 2) after finishing the element. The use of contact-type measurement devices (ruler & tape measures) determine the achievable precision and type of measurements. Complex mold geometries limit their use. The permissible geometric discrepancies between designed and built vary with the type of building elements. Façade elements require relatively high precision, especially in the position of mounting points (aim: functioning attachment), and for edges of elements (aim: aesthetically pleasing gap dimensions and alignment with adjacent elements). The tolerances for length are: +-3mm for elements < 3m, +-5mm for elements < 9m, and 8mm for linear straightness (Kim et al. 2020). Others quality aspects, like straightness and squareness, have higher permissible tolerances and are in practice hardly measured.

The above workflow shows a disconnection between the physical production and the digital element specification since the whole manufacturing process relies on the moldmaker's execution of 2d technical drawings.



Figure 1
Mold building and mold casting, both specialized hand craft processes

STATE OF THE ART IN PRODUCTION FEEDBACK

The quality of the finished GFRC element is dependent upon the quality achieved in each of the steps from design to fabrication. According to the NRC (National Research Council 1995) the ability to produce quality products hinges on four key competencies: modeling of process form and precision levels, design tolerancing of parts and products, selecting production processes that match part specifications, and applying quantitative measurement methods for inspection and process control. The first two-process modeling and design tolerancing-are of primary importance and drive the second two. The first three take place pre-production, while inspection can only take place during and post production.

The field of manufacturing engineering has developed multiple approaches to ensure and improve quality and efficiency in production. A crucial element is early and repeated testing and feedback at each stage of development and production of elements (Matisoff 1986) . This feedback can be organised within a single organisational level of production (Horizontal Control Loop) or between multiple organisational levels (Vertical Control Loops) (Schmitt et al. 2011). Horizontal Control describes how one process is being controlled and how this control loop interacts with other control loops (e.g. end-of-line inspection on shop floor level) Vertical Control describes a multi-scalar approach of how control loops of lower levels are monitored, controlled and designed (e.g. management assessment of business processes). The need for higher quality production methods, which can adapt quickly to changes in production and are efficient at minimal batch sizes - as in the call for an Industry 4.0 (Kagermann 2015) - has led to an extension of thinking about quality control in industry. This expands consideration to include 1) the organisational level, where feedback is extended to span customer, production and design in Closed-Loop Quality Systems [4], 2) temporal aspects, with an urge for continuous and real time feedback on human and machine processes (Even-Chaim 2019) and

3) the further development of flexible manufacturing systems (De Toni & Tonchia 1998). These last items resonate with research in architectural construction exploring online feedback systems for robotic manufacturing with (Sutjipto et al. 2018) or without humans in the loop (Nicholas et al. 2017).

The triggering moment for feedback in any quality loop is the act of data generation, the registration of the measured data to a reference model and the analysis and feedback of deviations between these (Martin Tamke et al. 2016).A vast array of large-scale metrology technology exists for the capturing of dimensional data(Peggs et al. 2009). Ìn the context of GFRC manufacturing, this list is narrowed down to fast, contactless and mobile techniques 3d Laser Scanner and Digital Photogrammetry. This is due to the specifics of GFRC production:

- Large dimensions, weight, complex geometry with few geometric features, undercuts, narrow spaces and uniform color of molds and GFRC elements
- Small batch size of production runs, which requires frequent 3d capture
- Dusty production environments - necessity to move gear gear off site, when not in use
- Financial constraints, require measurements to be carried out by production floor personnel (need for ease of use and high automatisation)

Both Laser Scanners and Digital Photogrammetry provide sufficient precision to match the minimal production tolerances of casting (1mm) (National Research Council 1995) and the required tolerances in the construction industry (M. Tamke et al. 2014). Laser Scanning is used in research for control of surface qualities in concrete production (Puri & Turkan, 2016; Wang et al. 2016) and onsite control of rebar and concrete formwork (Kim et al. 2020).

Once scan data is generated, it needs to be processed to enable dimensional feedback. However related industrial products [6,7], are generic tools made for use in mechanical industries. In order to achieve the required precision, these products must be operated by specialists who adapt the measurement to the specific object, postprocess data and execute deviation control in separate specialist software packages. And while technological developments allow for a high degree of automatisation in the registration of large sets of point cloud data (Chen & Medioni 1992; Ochmann et al. 2014; Schnabel et al. 2007), the automated analysis of dimensional deviations remains a field of research (Martin Tamke et al. 2016).

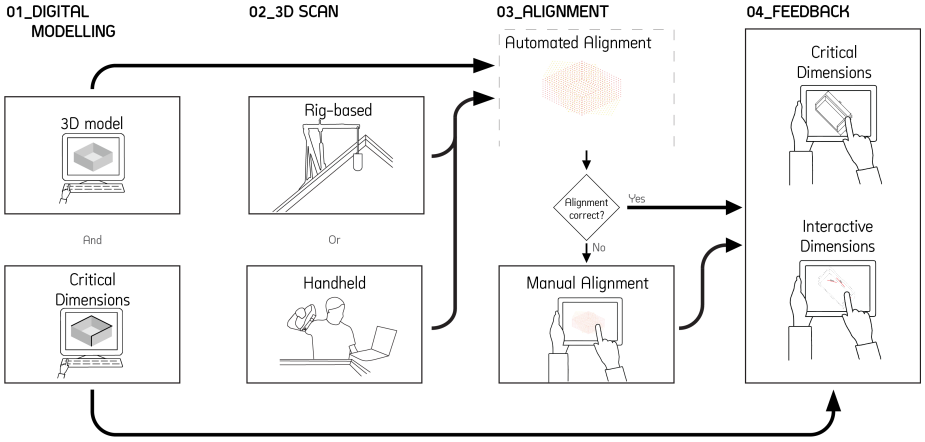
METHODS

In this paper we present a new, prototypical framework, termed Precision Partner - that enables the craft practice of mold-makers by establishing an automated connection to precise, digital information. In the previous section we have established the requirements for this novel approach to horizontal production feedback.

The workflow is established through software and hardware prototyping in direct dialogue and discussion with industry partners. We also develop a larger research methodology for industrial research collaboration, concerning real-case data gathering, quick Design/Development/Testing and feedback and engagement with the craftsmen and engineers to guide development and testing. Each software and hardware implementation is evaluated for its independent functionality and also for fit into the Precision Partner framework as a whole.

The development of the framework (Fig. 2) is guided by and tested against three main criteria. **First, the Precision Partner framework needs to be open-ended**, where components can be continuously added. This reflects the high level of variability in GFRC manufacture. A digital baseline approach, to handle incoming project information and generate construction information, as well as set the stage for registration and deviation feedback, is at the core of the framework. **Second, the framework necessitates an easy integration with existing production pipelines without too much disruption for the business**. For instance, differently from industrial product manufacturing, GFRC molds are made in

Figure 2
Relationship between the digital components and the human activities in the Precision Partner. Variability of the types of human interaction - station based, physical (3d scanning) and on the fly.



an adaptive production context, where the factory facilities provide a flexible production sequence, rather than a standard production chain, depending on the size and amount of elements to be produced. This is why we develop multiple flexible scanning scenarios. Finally and most importantly, **the framework needs to be accessible to a workforce without specialized computer skills**, since it will be used by shop-floor craftsmen. We design a simple interactive UI for on-the-fly dimension check, where the data processing is fully automated, allowing the craftsman to easily assess the quality of a component and react to any deficiencies. We expand upon the technicalities of each of these three development criterions in the sections below.

Digital modelling

The core of our production feedback lies in being able to compare the real element, to it's digital representation. For that, we require a standard set of outputs: The geometry of the mold or casted element in ply, with the geometry oriented orthogonally in XYZ space, as well as a list of the critical dimensions that need to be checked, such as height, length, width and thickness. We accommodate for element variability while ensuring a baseline quality. In fact, since element design and specification comes in different data formats from one project to the other, we develop input pipelines specific to different 3d model formats. However, when the GRC element is rectilinear and geometrically simple the input data from the client is provided as 2d drawing. As these are not fitting to a 3d approach, they require heavy manual editing in order to create consistent quality and labeling. We therefore develop a parametric configurator. This produces from simple numerical inputs,both a 2d technical drawing set and a cutting list to build the mold, as well as the necessary 3d model of the element and the mold. This drastically increases design-to-production productivity and guarantees consistent data quality for the following operations. It also allows very easily to create partial digital models, to compare against partial digital scans during the mold-building process.

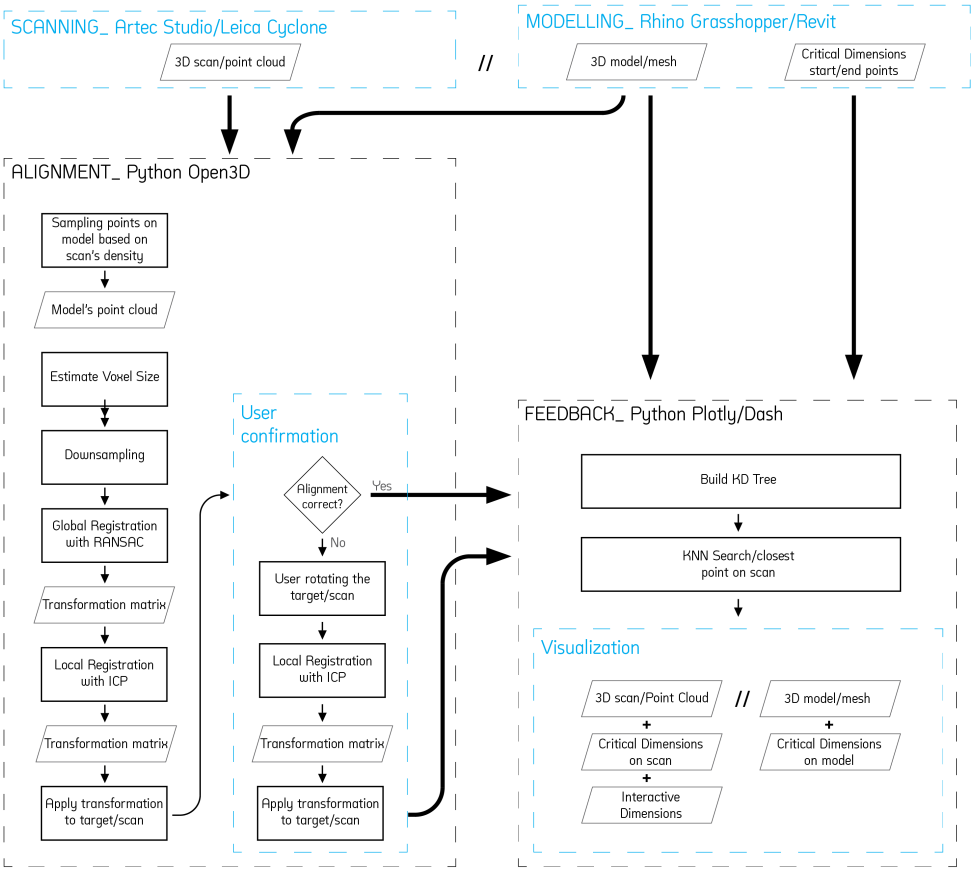


Figure 3
Back-end cloud processing workflows: processing of the acquired phyisical scan, alignement with digital model, comparison and feedback

Scanning Protocols

Horizontal feedback on the factory floor requires flexible scanning protocols in order to be minimally disruptive to the ongoing production chain. At different instances of the production, different parts of the mold can be scanned to check the quality of the built piece. Later, the whole mold can be checked prior to casting, and the element can be checked after unmolding. For that we develop two different scanning scenarios: partial or small scans and complete scans.

Partial Scans - Handheld. Handheld scanners operate through structured light. By shooting up to 16 frames per second, they are able to reconstruct the surface they are scanning on the fly using photogrammetry. Their use by craftsmen requires little training. Scans can be made of specific edges, or specific mounting points, which then feed into the registration workflows described below. A craftsman can also use printed trackers to check for a specific dimension of interest, that could be hard to measure

with a ruler for example. We envision that these scans would be used for fast intermediate checks during the mould building process.

Complete Scans - Rig Based. Lidar scanner.capture large areas quickly. This allows for a complete scan of the mold during manufacturing or for a final quality control. We build a rig with a suspended 3d scanner, which can be operated by a craftsman and positioned over the mould. The data is then passed to the registration workflows we describe below.

Cloud Processing Workflows

In order to give feedback on dimensions, the entire or partial scan, must be registered to its 3d model. We develop a workflow (Fig.3) for automatic registration using the Open3d library for Python (Zhou et al. 2018). The workflow uses a combination of RANSAC (Schnabel et al. 2007) algorithm for global registration and ICP (Chen and Medioni 1992) We prepare the point clouds of the scans through noise removal using nearest neighbour algorithms. We subsequently analyse the density of the scans and generate a point cloud from the 3d model with similar density, as this improves the success rate of the registration drastically. Since Ransac relies on an initial random seed, the results are previewed in the UI for confirmation. In the case of miss-alignment (currently at 5% of cases, but protocols are being developed to improve this) we developed a fallback manually-aided alignment based on matching the bounding boxes through simple slider-based operator inputs. The transformation matrix is applied to the scanned cloud to match it to the 3d model, allowing critical measures to be calculated on the scanned cloud using nearest neighbour search based on the baseline model. These workflows are all computed in the back-end and need under 1 minute computing time.

UI for precision feedback

To guarantee accessibility of this tool, the whole workflow is wrapped into a simple User Interface (UI) built using Dash library for python [8]. The UI runs on a web browser, and is built for interactive queries. At startup, the operator selects the project folder, in which cloud scan and 3d model are saved. The tool displays the pre-defined dimensions for both the scan and the model, and flags dimensions that are beyond tolerance. The operator can then use the spot-check feature to add additional dimensions, and better diagnose the problem.

WORKFLOW EVALUATION BALCONY CASE STUDY

In order to evaluate the performance of our workflow, we test it against a recurring type of element at our industry partner: The balcony panel type. These present a rectilinear base with edges on 3 sides. The case study in question measures 2554x1297 mm and is 170mm deep with a 12mm thickness. The edges are at 90 degree with respect to the base, but present a chamfer feature. The panel requires a 3x6 grid of metallic mounts. The parametric configurator generates a 2d technical drawing set, as well as mold cutting list for the mold builders. It also generates a 3d model of the element, and a 3d model of the mold (Fig. 4). We export these models and the critical dimensions to the Open3d workflow. The element mold is scanned both entirely and partially (Fig 5). Scans are processed as described in the section above. We generate a model point cloud consisting of approx. 3.5million points to match an approx. 4 million point scan. We test two registration methods on both partial and full scans, a fully automated one using Ransac and ICP and a manually augmented one (Fig 6). The latter utilizes trackers to create two sets of correspondence points. This is accomplished through the calculation of distances between the three points for each set and the forming of a sequence (e.g. p1 corresponds to p1', p2 corresponds to p2', etc.). After that, a transformation matrix is created in order to match the two point sets and it's applied to the scan. Through this the scan and the model are roughly registered. Finally, ICP is used for local refinement. The critical dimensions are computed on the UI (Fig 7) and the operator can visualize and interact with them.

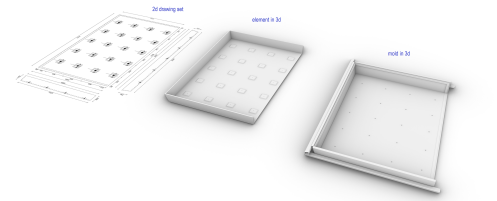


Figure 4
Outputs of the mold configurator parametric algorithm

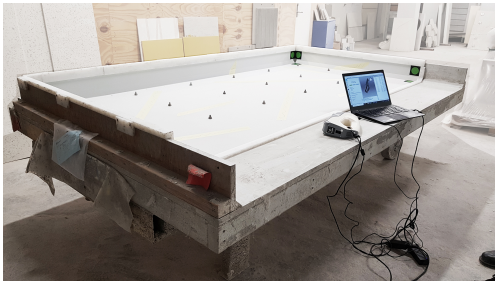


Figure 5
Scanning of a balcony type mold using Artec3d Eva handheld scanner

DISCUSSION

The conducted experiments have revealed that the relationship between 3d scan and 3d model is the cornerstone of the workflow. In order to be able to register the scans to its 3d model, they need to be of similar nature. We have found that the geometric features of the element have a big impact on the success of the registration algorithm. Full automation is successful when there is a 70-100% correspondence between the point clouds. Partial scans with less than 50% correspondence will require partial models to be registered to. This is possible when the model is generated from the parametric configurator. While being a theoretical use case this would allow to produce assembly sequence models, to be able to check at steps in the mold building process.

CONCLUSIONS AND FUTURE WORK

In this paper, we have described research towards a new framework for factory floor quality control that considers the human in the loop. The framework introduces Industry 4.0 technologies to enhance skilled craftsmen in the making of molds through the provision of horizontal feedback. It allows a craftsman to assess the geometric accuracy of their work against a digital model on the factory floor, through 3d scanning-based feedback of either pre or user-defined dimensions and deviations. This framework is tested through the case of a balcony element, where correspondence is found to be a key factor in successful automation.

As the paper details, the seemingly straightforward activity of gaining dimensional feedback via 3d scanning is found to require a wider, coordinated set of digital workflows. To achieve these for the high level of geometric complexity and variety that characterizes GFRC production, and with non-expert users rather than a skilled on-site technician, requires that the framework be both automated and open ended.

Automation is used to ensure the relationship between 3d scan and 3d model. An automated mold configurator tool is developed to enable a baseline for comparison between 3d model and point-cloud. A fully automated headless workflow correlates 3d scans made by either handheld or rig based scanning tools against a 3d model, using RANSAC and ICP algorithms. If needed, alternative workflows offer user input.

Open Endedness is achieved by constructing the framework as components. This enables different input pipelines, different modes of 3d scanning best suited to different mold geometries, extents of scanning, and registration either during or at the end of the making process. Here, further work is currently being undertaken to better enable the use of partial scans through specification of partial models, as a means to address the issue of correspondence in the registration of in-progress work. The question of when feedback can occur can be linked to the question of how feedback can be provided. Here, future work is planned to extend the framework to incorporate visual spatial feedback through overlay of point-cloud, model, deviation information and real world data through augmented reality.

Figure 6
Results of two registration processes of 3d model and scan: Upper two images - Fully automated RANSAC and ICP. Down two images - Manually augmented Correspondence points and ICP.

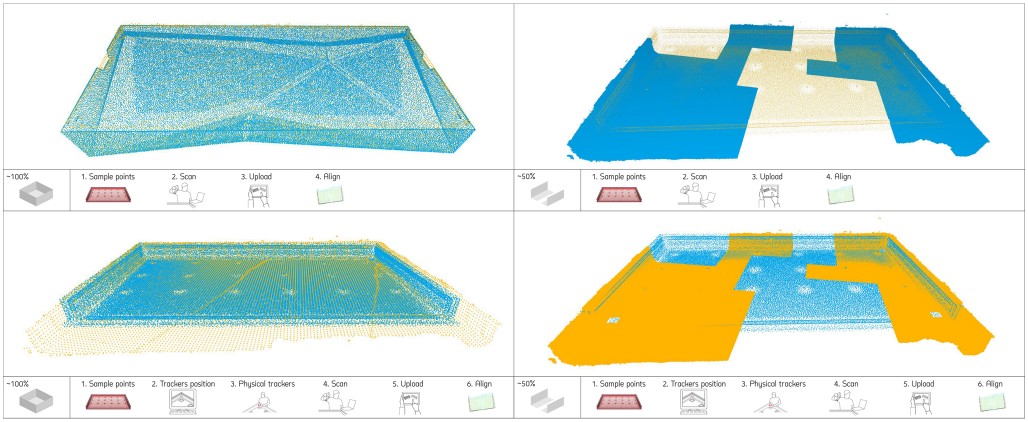
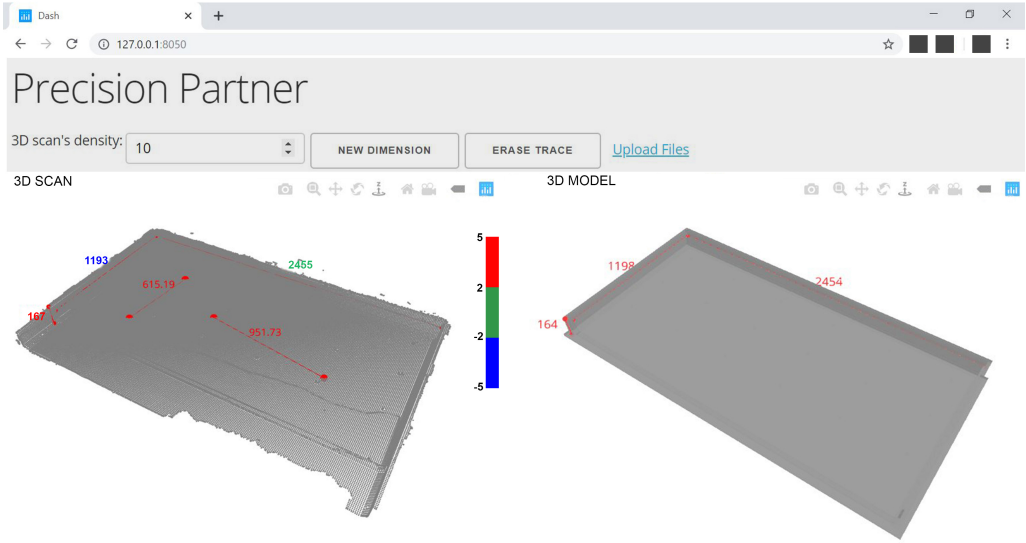


Figure 7
Our UI displayed on a web browser



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